Nathan Ma

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Education

University of Waterloo Sep 2024 - Present
BASc, Mechanical Engineering, Option in Artificial Intelligence *GPA: 4.0*

Experience

Toyota Motor Manufacturing Canada | *Weld Engineering Analyst Co-op* Jan 2026 - Apr 2026

- Led a TBP-driven **continuous improvement** project targeting shell-body sealer **vision system** downtime, reducing Coherix-related equipment fault time by **70%**. Applied structured problem-solving methodology to identify root causes and implemented corrective actions, presenting results to senior management and directly supporting North Weld line throughput improvement.
- Directed the integration of a **torque gun system** to accommodate two vehicle models with different requirements. Coordinated between production, maintenance, contractors, and vendors while overseeing weekend trial runs and commissioning activities involving PLC logic and system validation.
- Audited and optimized robot **weld schedules** to reduce weld spatter, eliminate weld separation defects, and decrease welding tip replacement frequency, contributing to estimated annual cost and material savings.
- Conducted time studies for over-cycle processes. Implemented improvements that reduced unnecessary robot movement and increased process catch-up capability for the process.

General Motors | *Industrial Engineering Co-op* May 2025 - Aug 2025

- Led a **continuous improvement** initiative focused on eliminating production inefficiencies through targeted **root cause analysis**, collaborating with **cross-functional teams** to optimize line balance. Successfully reduced job cycle times by up to **11.8%** and eliminated significant line downtime, resulting in an increased throughput of Chevrolet Silverado pickup trucks.
- Conducted **time studies** and redesigned standardized work across **30+** stations, utilizing **lean manufacturing** principles to identify wasted movement and ergonomic risks, resulting in noticeable cycle time improvements and reduced ergonomic strain.
- Updated **10,000+ sqft** of factory layout using **AutoCAD**, streamlining cross-functional operations by delivering precise, standardized floor plans that enhanced coordination and efficiency across departments.
- Developed practical understanding of **DFMA** by attending current and future product design review meetings and observing how part design decisions influenced cost, ergonomics, and error-proofing on the plant floor—connecting design intent with real-world assembly outcomes.
- Developed an **Excel-based** readiness document mapping job elements to their associated option codes, tooling, part numbers, and rack sizes, improving efficiency of plant floor changeovers.

Projects

Autonomous Quadruped | *SolidWorks, 3D-Printing, Soldering, KiCad, Embedded Programming*

- Developing a **12-DOF quadruped** with hierarchical control architecture using custom 12-channel **STM32** driver for low-level servo control, **Raspberry Pi** for high-level **perception**, **mapping**, and **motion planning**, with **LiDAR** for environmental sensing.

Internal Cycloidal Actuator | *SolidWorks, 3D-Printing, Soldering*

- Engineered a **compact 7:1 cycloidal actuator** for dynamic robotic joints at a low cost.
- Performed iterative design on gear and housing tolerances to reduce backlash while preserving backdrivability and achieving smooth, reliable motion.

Technical Skills

SolidWorks, Siemens NX, CATIA 3DX, ANSYS, AutoCAD, Excel, Python, C++, MATLAB, ROS2, Arduino, STM32, ESP32, Embedded Systems, Machining, Welding, GD&T, 3D Printing, Laser Cutting, Soldering, Fusion 360, Autodesk Inventor, FEA (Stress, Thermal, CFD), Topology Optimization, FMEA, Injection Molding, Rapid Prototyping (FDM, SLS, SLA), Silicone Casting, Leak Testing, Thermal Testing, Load Cell Integration, Mechanical Packaging, BOM Management, Materials Selection, Crimping, Wire Routing, Root Cause Analysis, Design of Experiments (DOE), Technical Documentation